

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AC1-210***

Madison 69 kV

June 2018

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: (1) Direct Connections, which are new facilities and/or facilities upgrades needed to connect the generator to the PJM network, and (2) Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances a generator interconnection may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the feasibility study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Madison Solar, LLC (Community Energy) proposes to install PJM Project #AC1-210, a 45.0 MW (31.0 MW Capacity) solar generating facility in Madison County, Ohio (see Figure 2). The point of interconnection will be a direct connection to AEP's Madison 69 kV substation (see Figure 1).

The requested in service date is June 30, 2018.

Attachment Facilities

Point of Interconnection (Madison 69 kV Substation)

To accommodate the interconnection at Madison 69 kV substation, Madison substation will have to be expanded to a new three (3) circuit breaker 69 kV substation physically configured in a breaker and half bus arrangement (see Figure 1). Installation of associated protection and control equipment, 69 kV line risers, SCADA, and 69 kV revenue metering will also be required. AEP reserves the right to specify the final acceptable configuration considering design practices, future expansion, and compliance requirements.

Station Work:

- Expand Madison 69 kV substation to a new three (3) circuit breaker 69 kV substation physically configured in a breaker and half bus arrangement (see Figure 1). Installation of associated protection and control equipment, 69 kV line risers, SCADA, and 69 kV revenue metering will also be required. (see Figure 1).
- **Estimated Station Cost: \$3,500,000**

Note: A supplemental area improvement project has been submitted to PJM that includes upgrades at Madison and the surrounding area. The scope of the station work at Madison for this interconnection will be affected by the supplemental project. Any changes to the scope of work will be addressed in the Facility Study.

Non-Direct Connection Cost Estimate

The total preliminary cost estimate for Non-Direct Connection work is given in the following tables below:

For AEP building Direct Connection cost estimates:

Description	Estimated Cost
69 kV Revenue Metering	\$200,000
Upgrade line protection and controls at the expanded Madison 69 kV substation.	\$850,000
Upgrade line protection and controls at the Harrison 69 kV substation	\$200,000

Description	Estimated Cost
to coordinate with the expanded Madison 69 kV substation.	
Upgrade line protection and controls at the Beatty Road 69 kV substation to coordinate with the expanded Madison 69 kV substation.	\$200,000
Total	\$1,450,000

Table 1

Interconnection Customer Requirements

It is understood that Community Energy is responsible for all costs associated with this interconnection. The cost of Community Energy's generating plant and the costs for the line connecting the generating plant to Community Energy's switching station are not included in this report; these are assumed to be Community Energy's responsibility.

The Generation Interconnection Agreement does not in or by itself establish a requirement for American Electric Power to provide power for consumption at the developer's facilities. A separate agreement may be reached with the local utility that provides service in the area to ensure that infrastructure is in place to meet this demand and proper metering equipment is installed. It is the responsibility of the developer to contact the local service provider to determine if a local service agreement is required.

Requirement from the PJM Open Access Transmission Tariff:

1. An Interconnection Customer entering the New Services Queue on or after October 1, 2012 with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.
2. The Interconnection Customer may be required to install and/or pay for metering as necessary to properly track real time output of the facility as well as installing metering which shall be used for billing purposes. See Section 8 of Appendix 2 to the Interconnection Service Agreement as well as Section 4 of PJM Manual 14D for additional information.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

AEP Requirements

The Interconnection Customer will be required to comply with all AEP Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements

may be found within the “Requirements for Connection of New Facilities or Changes to Existing Facilities Connected to the AEP Transmission System” document located at the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-aep/aep-interconnection-requirements.ashx>

Network Impacts

The Queue Project AC1-210 was evaluated as a 45.0 MW (Capacity 31.0 MW) injection into the Madison 69 kV substation in the AEP area. Project AC1-210 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AC1-210 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Base Case Used

Summer Peak Analysis – 2020 Case

Contingency Descriptions

The following contingencies resulted in overloads:

None

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

None

Short Circuit

(Summary of impacted circuit breakers)

New circuit breakers found to be over-duty:

None

Stability Analysis

No mitigations were found to be required

Affected System Analysis & Mitigation

LGEE Impacts:

None

MISO Impacts:

None

Duke, Progress & TVA Impacts:

None

OVEC Impacts:

None

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

Additional Limitations of Concern

AEP performed additional analysis for this interconnection request since PJM does not monitor AEP's 69 kV facilities. The local area study looked at the line loading and loads in the area to determine the effects of adding generation in the area. The study results showed load loss in the area will overload the Darbyville Tap – Texas Eastern 69 kV section of the Madison – Picway 69 kV circuit comprised of copper conductor (25 MVA rating). Due to the radial nature of the line, it cannot be rebuilt without first completing and closing the loop at Madison station in order to avoid extensive customer outages.

AEP has already submitted a supplemental project to PJM to mitigate these and other constraints in the area. The projected In-Service Date is 12/1/2019.

The maximum output from the AC1-210 plant will be capped at 25 MW to avoid overloading the limiting conductor section in the area until the supplemental project is completed.

New System Reinforcements

None

Schedule

It is anticipated that the time between receipt of executed agreements and Commercial Operation may range from 12 to 18 months if no line work is required. If line work is required, construction time would be between 24 to 36 months after signing an interconnection agreement.

Note: The time provided between anticipated normal completion of System Impact, Facilities Studies, subsequent execution of ISA and ICSA documents, and the proposed In-Service Date is shorter than usual and may be difficult to achieve.

Conclusion

Based upon the results of this System Impact Study, the construction Madison Solar's 45.0 MW (31.0 MW Capacity) solar generating facility (PJM Project #AC1-210) will require the following additional interconnection charges. This plan of service will interconnect the proposed solar generating facility in a manner that will provide operational reliability and flexibility to both the AEP system and the Madison Solar, LLC generating facility.

Cost Breakdown for Point of Interconnection (Rio-Lick 138 kV)			
Type of Network Upgrade	Network Upgrade Number	Description	Estimated Cost
Non-Direct Connection Network Upgrade	n5679	Expand Madison 69 kV Substation	\$3,500,000
	n5680	69 kV Revenue Metering	\$200,000
	n5681	Upgrade line protection and controls at the expanded Madison 69 kV substation.	\$850,000
	n5682	Upgrade line protection and controls at the Harrison 69 kV substation to coordinate with the expanded Madison 69 kV substation.	\$200,000
	n5683	Upgrade line protection and controls at the Beatty Road 69 kV substation to coordinate with the expanded Madison 69 kV substation.	\$200,000
		Total Estimated Cost for Project AC1-188	\$4,950,000

Table 2

The estimates are preliminary in nature, as they were determined without the benefit of detailed engineering studies. Final estimates will require an on-site review and coordination to determine final construction requirements.

Figure 1: Point of Interconnection (Madison 69 kV Substation)
Single-Line Diagram

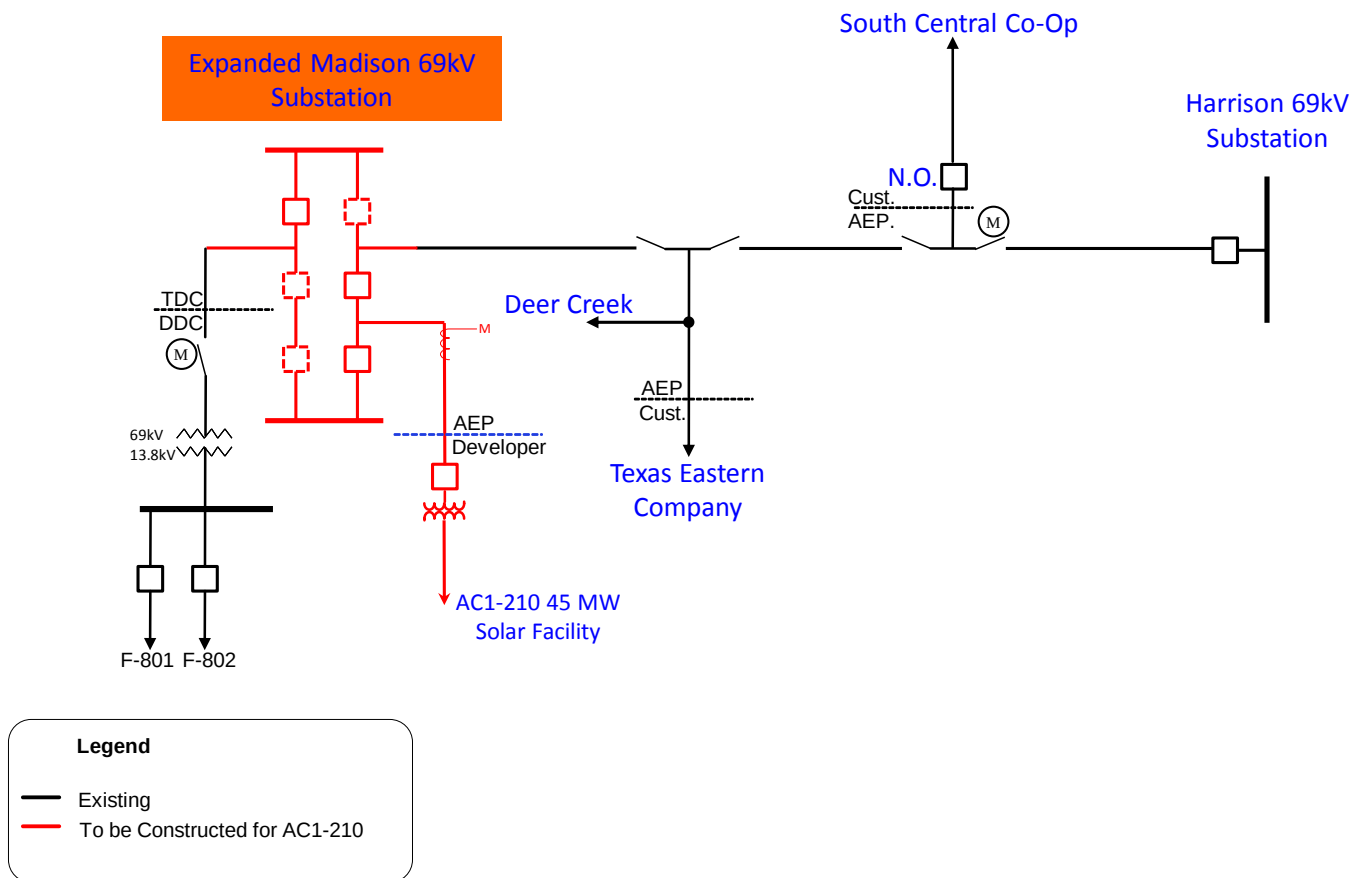


Figure 2: Point of Interconnection (Madison 69 kV Substation)

